Anti-Corruption Base Intelligence

Design Overview-V-1

Plantel: Centro Universitario de los Valles

Materia: Software Configuration Management

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Grado: Tercer semestre

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Design Overview

Design Overview with technologies
Architecture: Clean Architecture + Local-First
on the mobile device without a central database. This was designed to maintain control over sensitive
country data and to control who has access to the app.

Frontend: React, JavaScript
Backend/Core: Rust for code obfuscation

```
     [ Device  ]
        |
        v
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|      User Interface (UI)     |   <-- React
|  - Main Dashboard            |
|  - Network Graphs & Timelines |
|  +-------------------------+   |   <-- "API Manager" Section
|  | Interactive API Diagram |   |       - Manual Add/Edit APIs
|  | (vis.js or D3.js)       |   |       - Drag-and-Drop to Connect & Reorganize
|  | - Nodes: Configured APIs|   |       - Test Button & Status Indicators
|  | - Edges: Flow to Modules|   |
|  +-------------------------+   |
+-------------------------------+
         ↑↓ (Tauri IPC / Commands) -It helps protect the application between the interface and the core logic.
         |
+-------------------------------+
|      Rust Core (Backend)     |   <-- Obfuscated business logic
|                              |
|  - Domain / Entities         |   <-- Core objects: Officials, Companies, Projects, Relationships, etc.
|  - Use Cases (Modules 1–5)   |   <-- Data Listener, Network Analysis, Process Tracking, RTL, Time Paradox
|  +-------------------------+   |   <-- API Configuration Service - For User "Detectives for Line Investigation Cases"
|  | API Config Manager      |   |       - CRUD for manual APIs
|  | - Store & Parse Configs |   |       - Dynamic Parsing (JSONPath)
|  | - Validation & Testing  |   |       - Resilience (retries, fallbacks)
|  +-------------------------+   |
+-------------------------------+
         ↑↓ (Local  Channels)
         |
```

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|      Infrastructure Layer    |   <-- All local, encrypted
|  - Encrypted Local DB         |   <-- SQLCipher -It is a solution for storing data locally in an
encrypted manner.
|  - Immutable Chain of Custody |   <-- Hashes + certified timestamps
|  - E2EE & libsodium           |    -End-to-End Encryption and libsodium cryptographic functions
|  - Tor Integration (arti)     |   <-- Automatic IP rotation, padding, dummy traffic
|  - VPN Support                |
|  - Resilient API Client       |   <-- Now uses user-defined APIs, for the anonymity of the
Investigators
|  - Anti-Forensic Features:    |
|    - Sandbox / Dump Detection |
|    - Screen Block             |
|    - Self-Destruct            |
|    - Inverse Honeypot (fake data)|
|    - Code Obfuscation         |
+-------------------------------+
         ↑↓ (Foreground / Limited Background) -This means that access to external sources
(public APIs, logs, etc.) only occurs when the app is active.
         |
[ External Sources ] <-- Only when app is active or in foreground service
  - APIs (configured manually by detective)
```